Fundamental Theorem of Galois Theory.

Thm. Let K/F be a Galois extension with Galois group $Gal(K/F)$. The map

$$G_{K/F}: \{E \mid F \leq E \leq K\} \leftrightarrow \{H \mid \{id\} \leq H \leq Gal(K/F)\}$$

$$: \quad E \mapsto Aut(K/E), \quad K^H \hookleftarrow H$$

is a bijection. Moreover, $G_{K/F}$ satisfies the following properties:
Given $(E_1, H_1), (E_2, H_2) \in G_{K/F}$,

1) $E_1 \leq E_2 \Leftrightarrow H_2 \leq H_1$

Given $(E, H) \in G_{K/F}$,

2) $[K:E] = |H|$ and $[E:F] = |G:H|$

3) K/E is always Galois with $Gal(K/E) = H$.

4) E/F is normal extension $\Leftrightarrow H \trianglelefteq Gal(K/F)$ is normal subgroup.

Proof of the theorem.